

```
import pandas as pd
import numpy as np

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns

# Machine Learning
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import BaggingClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
# Load dataset
data = pd.read_csv("Lung Cancer Dataset.csv")

# Display first 5 rows
data.head()
```

	AGE	GENDER	SMOKING	FINGER_DISCOLORATION	MENTAL_STRESS	EXPOSURE_TO_POLLUTION	LONG_TERM_ILLNESS	ENERGY_LEVEL	IMMUNE
0	68	1	1	1	1	1	0	57.831178	
1	81	1	1	0	0	1	1	47.694835	
2	58	1	1	0	0	0	0	59.577435	
3	44	0	1	0	1	1	0	59.785767	
4	72	0	1	1	1	1	1	59.733941	

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 18 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   AGE                                    5000 non-null   int64
 1   GENDER                                5000 non-null   int64
 2   SMOKING                               5000 non-null   int64
 3   FINGER_DISCOLORATION                 5000 non-null   int64
 4   MENTAL_STRESS                        5000 non-null   int64
 5   EXPOSURE_TO_POLLUTION               5000 non-null   int64
 6   LONG_TERM_ILLNESS                   5000 non-null   int64
 7   ENERGY_LEVEL                       5000 non-null   float64
 8   IMMUNE_WEAKNESS                     5000 non-null   int64
 9   BREATHING_ISSUE                     5000 non-null   int64
10   ALCOHOL_CONSUMPTION                 5000 non-null   int64
11   THROAT_DISCOMFORT                  5000 non-null   int64
12   OXYGEN_SATURATION                   5000 non-null   float64
13   CHEST_TIGHTNESS                     5000 non-null   int64
14   FAMILY_HISTORY                      5000 non-null   int64
15   SMOKING_FAMILY_HISTORY              5000 non-null   int64
16   STRESS_IMMUNE                       5000 non-null   int64
17   PULMONARY_DISEASE                   5000 non-null   object
dtypes: float64(2), int64(15), object(1)
memory usage: 703.3+ KB
```

```
data.isnull().sum()
```

	0
AGE	0
GENDER	0
SMOKING	0
FINGER_DISCOLORATION	0
MENTAL_STRESS	0
EXPOSURE_TO_POLLUTION	0
LONG_TERM_ILLNESS	0
ENERGY_LEVEL	0
IMMUNE_WEAKNESS	0
BREATHING_ISSUE	0
ALCOHOL_CONSUMPTION	0
THROAT_DISCOMFORT	0
OXYGEN_SATURATION	0
CHEST_TIGHTNESS	0
FAMILY_HISTORY	0
SMOKING_FAMILY_HISTORY	0
STRESS_IMMUNE	0
PULMONARY_DISEASE	0

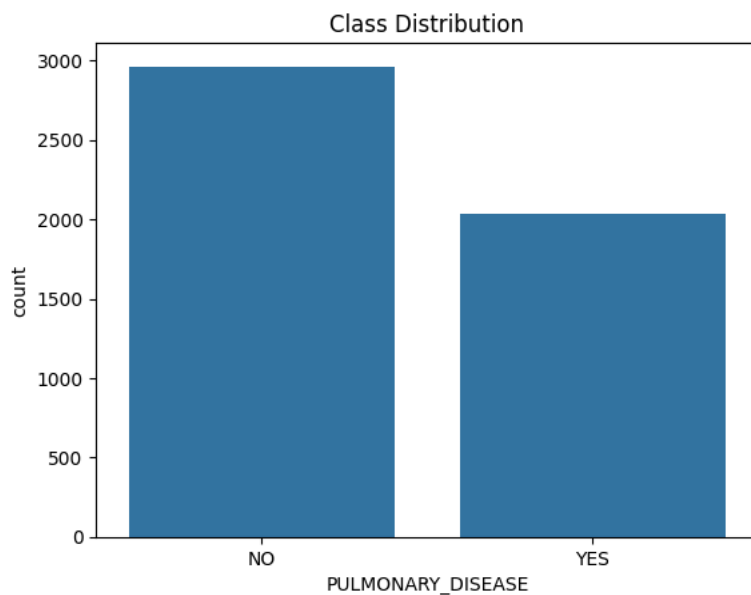
dtype: int64

```
data['PULMONARY_DISEASE'].value_counts()
```

	count
PULMONARY_DISEASE	
NO	2963
YES	2037

dtype: int64

```
sns.countplot(x='PULMONARY_DISEASE', data=data)
plt.title("Class Distribution")
plt.show()
```



```
# Convert categorical columns to numeric
data = pd.get_dummies(data, drop_first=True)
```

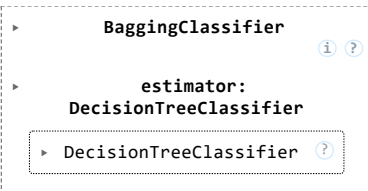
```
X = data.drop('BREATHING_ISSUE', axis=1)
y = data['STRESS_IMMUNE']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)
```

```
base_model = DecisionTreeClassifier(random_state=42)
```

```
bagging_model = BaggingClassifier(
    estimator=base_model,
    n_estimators=100,
    max_samples=0.8,
    bootstrap=True,
    random_state=42,
    n_jobs=-1)
```

```
bagging_model.fit(X_train, y_train)
```



```
y_train_pred = bagging_model.predict(X_train)
y_test_pred = bagging_model.predict(X_test)
```

```
train_accuracy = accuracy_score(y_train, y_train_pred)
test_accuracy = accuracy_score(y_test, y_test_pred)

print("Training Accuracy:", train_accuracy)
print("Testing Accuracy:", test_accuracy)
```

```
Training Accuracy: 1.0
Testing Accuracy: 1.0
```

```
print("Classification Report:\n")
print(classification_report(y_test, y_test_pred))
```

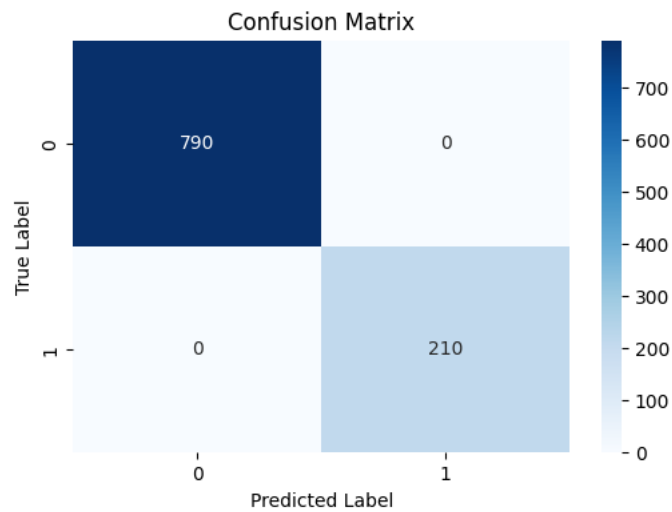
```
Classification Report:
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	790
1	1.00	1.00	1.00	210
accuracy			1.00	1000
macro avg	1.00	1.00	1.00	1000
weighted avg	1.00	1.00	1.00	1000

```
cm = confusion_matrix(y_test, y_test_pred)
cm
```

```
array([[790,  0],
       [ 0, 210]])
```

```
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Confusion Matrix")
plt.show()
```



```
if train_accuracy > test_accuracy:
    print("Model may be slightly overfitting.")
else:
    print("Model generalizes well.")
```

Model generalizes well.

```
results = []

for n in [10, 50, 100, 200]:
    for depth in [None, 5, 10]:
        model = BaggingClassifier(
            estimator=DecisionTreeClassifier(max_depth=depth),
            n_estimators=n,
            max_samples=0.8,
            random_state=42,
            n_jobs=-1
        )
        model.fit(X_train, y_train)
        y_pred = model.predict(X_test)
        acc = accuracy_score(y_test, y_pred)
        results.append((n, depth, acc))

results_df = pd.DataFrame(results, columns=['Trees', 'Max Depth', 'Test Accuracy'])
results_df.sort_values(by='Test Accuracy', ascending=False)
```

	Trees	Max Depth	Test Accuracy
0	10	NaN	1.0
1	10	5.0	1.0
2	10	10.0	1.0
3	50	NaN	1.0
4	50	5.0	1.0
5	50	10.0	1.0
6	100	NaN	1.0
7	100	5.0	1.0
8	100	10.0	1.0
9	200	NaN	1.0
10	200	5.0	1.0
11	200	10.0	1.0

```
best_model = results_df.loc[results_df['Test Accuracy'].idxmax()]
print("Best Model Configuration:")
print(best_model)
```

Best Model Configuration:

Trees 10.0

Max Depth NaN

Test Accuracy 1.0

Name: 0, dtype: float64

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